भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 306: Machine Learning Lab
Evaluation Assignment 3

Instructions: This assignment is for evaluation and marks will be awarded. You need to complete execution by 12 PM. The saved results folder (consisting .csv files, plots, word file) has to be submitted through a Google form, which will be shared by Teaching Assistant.

1. A financial institution wants to identify groups of similar credit card transactions and detect unusual transactions that may indicate fraudulent activities. Consider the Credit Card Fraud Detection dataset available on Kaggle.

   The dataset contains the following attributes:

   - Time: Time elapsed between a transaction and the first transaction.
   - V1–V28: Anonymized numerical features obtained using PCA.
   - Amount: Transaction amount.
   - Class: Transaction label, where 0 represents a legitimate transaction and 1 represents a fraudulent transaction.

   Dataset: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

   Task: Apply a suitable clustering algorithm to group transactions with similar characteristics and identify unusual observations that may indicate fraudulent activities.

   Perform the following tasks:

   (a) Read the dataset and report its dimensions, attribute names, descriptive statistics, and missing values. Report the number and percentage of legitimate and fraudulent transactions.

   (b) Extract V1-V28 and Amount as input features. Exclude Class and Time from clustering. Standardize the selected features using Z-score standardization (if you think it necessary). Explain why standardization is necessary.

   (c) Implement a suitable clustering algorithm to group transactions with similar characteristics and identify unusual observations.

   (d) Report the number of clusters, number of unusual observations, and percentage of unusual observations for each configuration, where applicable. Analyze how parameter changes affect cluster formation and unusual-observation detection.

   (e) Visualize the clustering results using a two-dimensional scatter plot of V1 and V2. Use different colors for clusters and a distinct marker for unusual observations, where applicable. Also visualize the actual legitimate and fraudulent transactions. Discuss the observed patterns.

(f) Compare the identified unusual observations with the actual `Class` labels. Compute precision, recall, and F1-score for fraud detection wherever anomaly predictions can be obtained. Explain why unusual observations are not necessarily fraudulent transactions.

(g) Compute the Silhouette Score for each configuration where mathematically valid. Exclude explicitly identified noise points before evaluation. Report the scores and explain why the Silhouette Score may not be applicable to every configuration.

(h) Compare the five configurations using cluster quality, unusual-observation detection, Silhouette Score, and fraud detection metrics, where applicable. Identify the most suitable configuration and justify your choice.

(i) Summarize the results in a Word document, including dataset description, preprocessing, selected clustering algorithm, parameter configurations, result tables, plots, evaluation metrics, and conclusions.